PROJECT REPORT

Detection and removal of adapter-contaminated reads

Oxford Nanopore sequencing · Summary reconstructed from supplied project documents

215,176

reads processed

61,495

reads passed · 28.6%

38.10 M

bases retained · 27.4%

Executive summary

The source report states that Cutadapt processed 215,176 reads, discarded 153,681 reads as trimmed, and retained 61,495 reads. The presentation includes six adapter-match length plots. Their numeric point tables and the raw FASTQ/POD5 inputs were not supplied, so this package reproduces the aggregate summary and preserves the source plots without digitizing them.

Scope statement

This is a reported-results summary, not an independent re-analysis or a validated read-level chimera classification. The documented --discard-trimmed option filters reads with configured matches; it does not split chimeric reads.

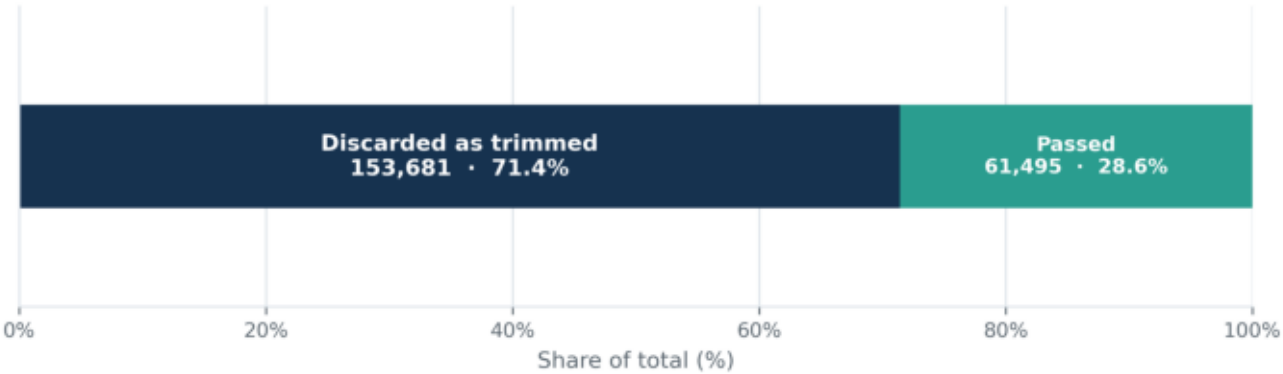
Source: NGSproject report pp. 7–18; final presentation slides 12–19.

Reported filtering outcomes

Both charts use only values transcribed from the source summary.

Reported read filtering outcome

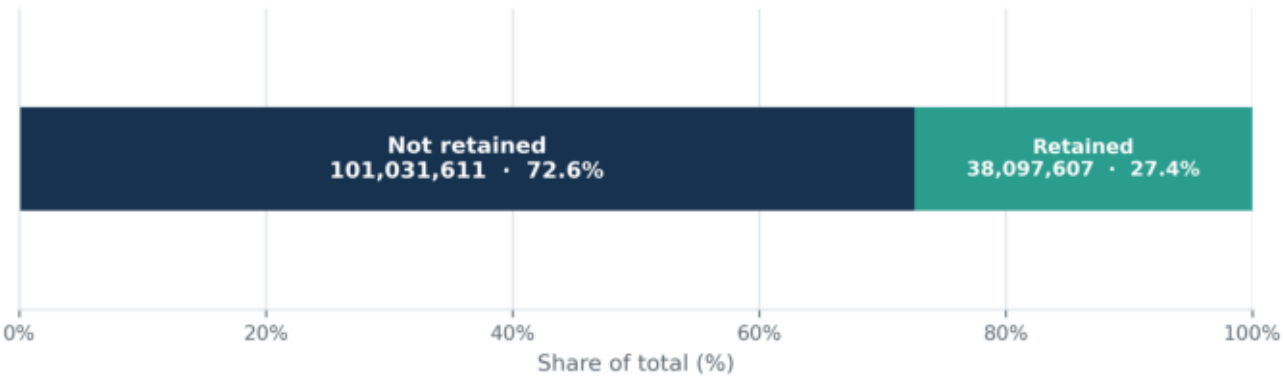
Reported total: 215,176



Source-reported aggregate; not independently recalculated from raw reads.

Reported base-pair retention

Reported total: 139,129,218



Source-reported aggregate; not independently recalculated from raw reads.

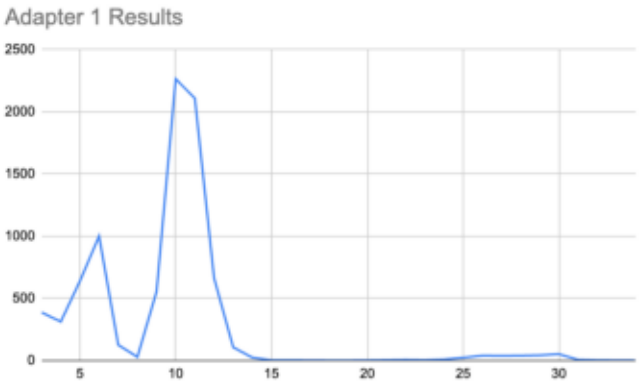
Derived from reported totals: 101,031,611 bp were not retained.

Adapter-match distributions

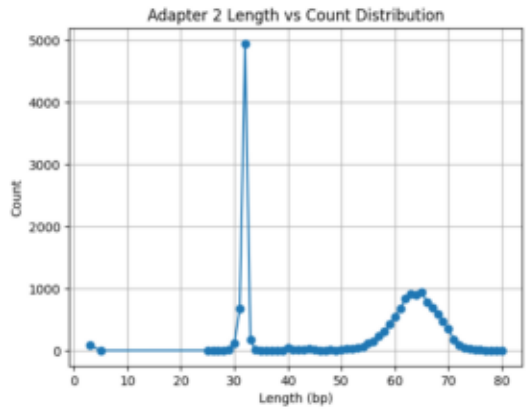
Six source plots from the presentation; not redrawn from point-level data.

Adapter-match length distributions from the supplied presentation

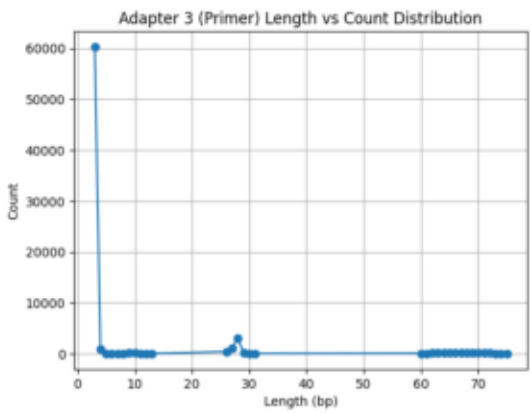
A1 · Native adapter



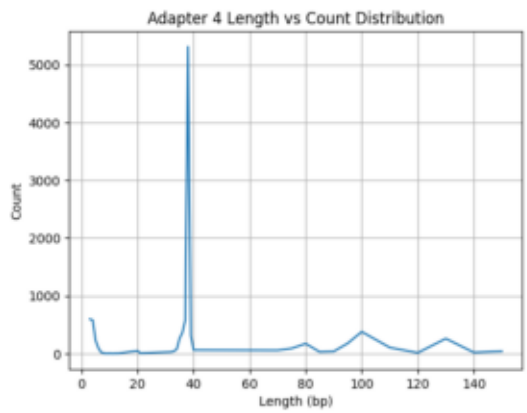
A2 · Barcode NB022



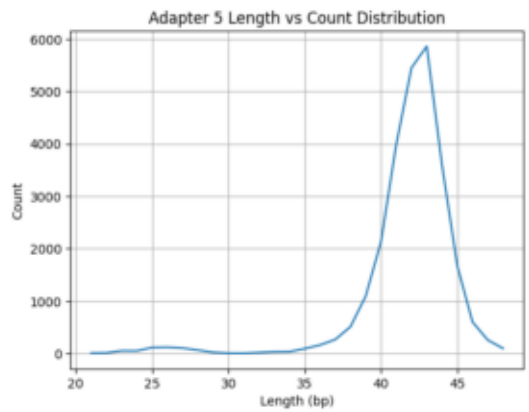
A3 · 27F primer



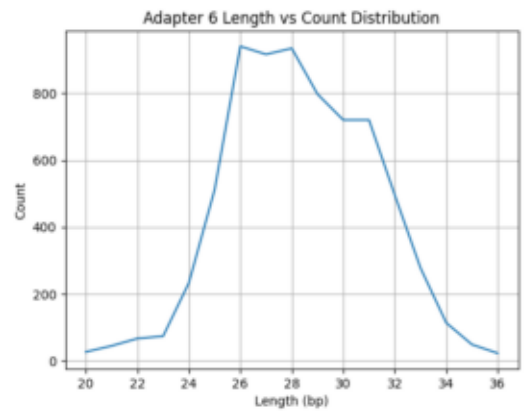
A4 · 1492R primer



A5 · Reverse barcode



A6 · Reverse native adapter



Source plots reproduced as supplied (slides 13–18); numeric point tables and read-level match coordinates were not provided.

Longer match-length patterns are interpreted in the source report as possible internal matches; read-level coordinates are needed to verify that interpretation.

Methods and interpretation

Workflow as reported

The project report states that MinKNOW high-accuracy basecalling was performed with adapter and barcode trimming disabled. The resulting FASTQ files were combined and processed with Cutadapt using six 5-prime/3-prime adapter models, reverse-complement handling, and --discard-trimmed.

What the filter means

The reported option discards reads that are trimmed against configured adapters and writes reads without a detected configured match. It is not equivalent to splitting a concatenated read into valid fragments, and it does not by itself classify a read as a PCR chimera.

What the plots show

The supplied slides show adapter-match length/count patterns. Per-adapter trim occurrences are transcribed where visible in the source log snapshots. Such counts may overlap across adapters and are not unique-read counts. The point tables and match positions were not supplied.

Evidence-based conclusion

The documented output supports a high rate of reads removed by the configured sequence-matching filter. The source report proposes internal adapters, concatenation, and PCR chimeras as explanations; those hypotheses require read-level sequence or alignment evidence before they can be confirmed.

Associated laboratory context

Separate evidence stream

A separate workflow document records extraction, gel electrophoresis, PCR, DNA quantification, and Nanopore library-preparation steps. The supplied files do not link these observations to the FASTQ dataset summarized in the NGS report.

Reported QC concerns

The workflow notes describe an early gel run with a maximum voltage of 33 V and poor separation, an unexpected band in the T1 no-template PCR control, uncertain quantification calculations, and no visible DNA bands on the final library gel.

Interpretation boundary

These are statements from the laboratory record, not independent validation. They should not be used to explain the Nanopore read-filtering totals until sample IDs and run provenance are linked. The source document lists possible causes but does not resolve them.

Public repository contents

The source PDFs and extracted text transcripts are intentionally omitted from this public repository. The included report, figures, CSVs, and scripts are the project deliverables selected for publication.

Reproducibility and next steps

Reproduce the summary package

Install requirements.txt, then run scripts/validate_metrics.py, scripts/generate_figures.py, scripts/generate_report.py, and python -m unittest discover -s tests -v. These steps validate and visualize transcribed aggregate values; they do not re-run the original read analysis.

Run the documented filter

Provide the original FASTQ and run scripts/run_documented_cutadapt.sh. Save the Cutadapt version, JSON report, full log, and sample manifest so that the reported counts can be reconciled.

Complete chimera validation

Export numeric adapter-match data and coordinates, predefine read-level criteria, and apply an amplicon-appropriate chimera classifier or split-read alignment. Then compare mapping or assembly metrics before and after filtering.

Inputs still needed

Raw FASTQ/POD5 reads, full Cutadapt log or JSON, the numeric distribution data, a reference/amplicon description, and sample IDs that connect wet-lab QC to sequencing records.